Huawei Access Node KR-ACC-41 QoS Congestion Event Resolution SOP

Node: KR-ACC-41 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_6912

Description

This Standard Operating Procedure outlines the steps for resolving a Major Severity (ALARM_6912) QoS Congestion Event on Huawei Access Node KR-ACC-41 located at KR Puram Access. The root cause of this failure domain is attributed to microbursts, interface tail drops, and QoS queue starvation. This issue is specific to this node due to the inherent design of the Huawei access node's QoS architecture, which is based on the DiffServ model. The DiffServ model uses a hierarchical structure to classify and schedule traffic, with the ingress node responsible for classifying traffic into different classes of service (CoS). If the ingress node is overwhelmed with traffic, it can lead to queue congestion, resulting in microbursts, interface tail drops, and QoS queue starvation. Additionally, the Huawei access node's QoS architecture relies heavily on the use of QoS queues, which can become congested if not properly configured. Furthermore, the node's congestion management mechanism is based on the Weighted Random Early Detection (WRED) algorithm, which can be triggered by high traffic loads, leading to further congestion and packet loss.

Pre-requisite Checks

- Verify that all redundant links are functioning properly and there are no issues with the network topology.

- Check the node's temperature and ensure it is within the recommended operating range.

- Verify that the node's power supply is stable and not experiencing any issues.

- Dispatch remote hands to KR Puram Access to confirm the issue and perform any necessary physical checks.

Potential Root Causes

Cause	Probability
Misconfigured QoS queues leading to congestion	High
Inadequate buffer sizes causing queue overflow	Medium
High traffic loads overwhelming the node's congestion management mechanism	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos queue, Expected Output: Queue Name Queue Type Queue Status Queue Depth Queue Threshold Queue Drop Queue Packet Loss, Description: This command displays the QoS queue configuration and status, helping to identify any misconfigured queues or inadequate buffer sizes.
```

```
Command: display qos queue statistics, Expected Output: Queue Name Packet Count Byte Count Drop Packet Count Drop Byte Count, Description: This command displays the QoS queue statistics, helping to identify any issues with queue overflow or congestion.

Command: display interface congestion, Expected Output: Interface Name Congestion Status Congestion Threshold Congestion Packet Loss, Description: This command displays the interface congestion status and statistics, helping to identify any issues with the node's congestion management mechanism.
```

Resolution Steps

Step 1: Step: Configure QoS queues to increase buffer sizes, Command: qos queue buffer-size , Description: Increase the buffer size of the QoS queues to prevent queue overflow and congestion.

Step 2: Step: Adjust QoS queue thresholds to prevent congestion, Command: qos queue threshold , Description: Adjust the QoS queue thresholds to prevent congestion and packet loss.

Step 3: Step: Verify that the node's congestion management mechanism is functioning properly, Command: display interface congestion, Description: Verify that the node's congestion management mechanism is functioning properly and not causing any issues.

Post-Check Verification

```
Command: display qos queue, Expected Output: Queue Name Queue Type Queue Status Queue Depth Queue Threshold Queue Drop Queue Packet Loss, Description: Verify that the QoS queue configuration and status are correct and within expected ranges.

Command: display qos queue statistics, Expected Output: Queue Name Packet Count Byte Count Drop Packet Count Drop Byte Count, Description: Verify that the QoS queue statistics are within expected ranges and not indicating any issues with queue overflow or congestion.

Command: display interface congestion, Expected Output: Interface Name Congestion Status Congestion Threshold Congestion Packet Loss, Description: Verify that the interface congestion status and statistics are within expected ranges and not indicating any issues with the node's congestion management mechanism.
```

Roll-back Procedures

- Step: Restore QoS queue buffer sizes to original values, Command: qos queue buffer-size , Description: Restore the QoS queue buffer sizes to their original values in case the fix fails.

- Step: Restore QoS queue thresholds to original values, Command: qos queue threshold , Description: Restore the QoS queue thresholds to their original values in case the fix fails.